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Assessment of two culturally competent Diabetes education methods: Individual versus Individual plus Group education in Canadian Portuguese adults with Type 2 Diabetes

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Objective. *To examine the impact of two culturally competent diabetes education methods, individual counselling and individual counselling in conjunction with group education, on nutrition adherence and glycemic control in Portuguese Canadian adults with type 2 diabetes over a three-month period.*

Design. *The Diabetes Education Centre is located in the urban multicultural city of Toronto, Ontario, Canada. We used a three-month randomized controlled trial design. Eligible Portuguese-speaking adults with type 2 diabetes were randomly assigned to receive either diabetes education counselling only (control group) or counselling in conjunction with group education (intervention group). Of the 61 patients who completed the study, 36 were in the counselling only and 25 in the counselling with group education intervention. We used a per-protocol analysis to examine the efficacy of the two educational approaches on nutrition adherence and glycemic control; paired t-tests to compare results within groups and analysis of covariance (ACOVA) to compare outcomes between groups adjusting for baseline measures. The Theory of Planned Behaviour was used to describe the behavioural mechanisms that influenced nutrition adherence.*

Results. *Attitudes, subjective norms, perceived behaviour control, and intentions towards nutrition adherence, self-reported nutrition adherence and glycemic control significantly*

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improved in both groups, over the three-month study period. Yet, those receiving individual counselling with group education showed greater improvement in all measures with the exception of glycemic control, where no significant difference was found between the two groups at three months.

Conclusions. *Our study findings provide preliminary evidence that culturally competent group education in conjunction with individual counselling may be more efficacious in shaping eating behaviours than individual counselling alone for Canadian Portuguese adults with type 2 diabetes. However, larger longitudinal studies are needed to determine the most efficacious education method to sustain long-term nutrition adherence and glycemic control.*

Keywords: Randomized Control Trial; Diabetes Education; Cultural Competency; Group Education; Portuguese Canadians

Introduction

Diabetes is a large and growing public health concern worldwide. The number of persons with diabetes, predominantly type 2 diabetes, has increased dramatically and is expected to grow substantially over the next several decades (King *et al*. 1998). Diabetes-related morbidity (i.e. retinopathy, neuropathy, nephropathy, cerebrovascular, peripheral vascular and cardiovascular diseases, immunosuppression and sexual dysfunction) and early mortality impose a heavy burden on those affected by diabetes (Stamler *et al*. 1993; Bertoni *et al*. 2002). In fact, it is amongst the five leading causes of death by disease in most countries (Zimmet 2000). To reduce or prolong the development of diabetes complications, researchers, health care providers and policy makers are seeking effective methods of education and management of diabetes.

An essential component in diabetes management is nutrition therapy (Franz *et al*. 1995). Glycated hemoglobin (HbA_{1c}) was reduced by an absolute 2% during the first three months with intensive diet therapy in newly diagnosed persons with type 2 diabetes (Eeley 1996) representing a potential risk reduction in microvascular and macrovascular complications (UK Prospective Diabetes Study Group 1998; UKPDS 1998b). Furthermore, nutrition therapy integrated as a part of diabetes self-management education (DSME) has been proven to be efficacious and cost-effective in improving blood pressure, serum lipids and glycemic control (Johnson & Valera 1995; Sheils *et al*. 1999; Miller *et al*. 2002). Despite the evidence that supports the benefits of DSME with a focus on nutrition therapy, limited information is available on the efficacy of DSME for minority groups.

For many years, minority groups have had the highest burden of diabetes (Carter *et al*. 1996). They have had limited access to and use of diabetes self-management resources and support (Bruce *et al*. 2003; Rothman *et al*. 2004; Strine *et al*. 2005), and overall, have poorer glycemic control (Carter *et al*. 1996; Koro *et al*. 2004).

Moreover, traditional DSME approaches for minority populations have been met with limited success due to low participation and retention rates (Bruce *et al*. 2003). Possible explanations for these low rates may include low levels of English literacy, a lack of culturally appropriate nutrition modification options and income constraints (Gohdes 1988). Moreover, minority groups whose primary language is not English may experience language discordance with their English-speaking care providers. Language discordance can lead to a decrease in patient comprehension, compliance and consequently poor clinical outcomes (Perez-Stable *et al*. 1997; Cooper *et al*. 2003b). Therefore, culturally competent programs that are efficacious in assisting minority populations to manage their diabetes are required to reduce language and cultural barriers, and to increase the appeal of DSME.

The goal of a culturally competent program is to develop culturally and linguistically appropriate services that function within the context of the cultural beliefs, behaviours and needs presented by an ethnic group (Anderson *et al*. 2003). Since each culture has distinct values and beliefs towards health, health behaviours and eating practices, culturally competent DSME programs are necessary to help individuals adopt dietary modifications relevant to their cultural preferences (Auslander *et al*. 2002).

While the data is insufficient for various cultural groups, the literature does suggest that DSME be delivered within small groups for individuals with diabetes from similar cultural backgrounds (Piette 1999). Group education produces several beneficial effects: interaction with and support from peers (Arseneau *et al*. 1994); opportunities to learn and gain motivation from each other through sharing experiences, beliefs and goals; and reducing feelings of fear and isolation (Payne 1995). These effects create an environment that helps people accept their disease and facilitate behaviour change (Arseneau *et al*. 1994; Cooper *et al*. 2003a). Furthermore, several randomized control trials (RCT) show that group education improves behaviour modifications, glycemic control and reduces the risks of cardiovascular disease (Raz *et al*. 1988; Trento *et al*. 2001, 2002; Rickheim *et al*. 2002). In addition, group education has also been observed to be economically effective (Trento *et al*. 2002). However, this literature does not represent studies conducted in culturally diverse minority groups.

The first objective of this pilot study was to determine the impact of two culturally competent education methods (individual counselling only and individual counselling in conjunction with group education) on nutrition adherence and glycemic control in Portuguese Canadian adults with type 2 diabetes. The second objective was to compare and explain differences in outcomes between the two groups using the Theory of Planned Behaviour. We hypothesized that all study participants would experience: (a) stronger intentions and a greater adherence to nutrition recommendations and (b) greater glycemic control after three months of starting the culturally competent program. We also hypothesized that participants receiving the intervention, individual counselling in conjunction with group education, would experience greater overall improvements ([a] and [b]) than participants in the control group, those receiving individual counselling alone.

Methods

Research Setting and Intervention

This study was conducted at the Toronto Western Hospital Diabetes Education Centre (DEC) located in an urban and culturally diverse area in Toronto, Ontario, Canada. In fact, it serves one of the world's most multicultural areas and the largest Portuguese-speaking population in Canada. The DEC core team consists of two endocrinologists, eight diabetes educators (four nurses and four dieticians) and a social worker. Additional professionals such as a physiotherapist, pharmacist and psychologist also participate in the group education program and patient consults. Referrals are regularly made to a chiropodist and ophthalmologist.

Culturally Competent Components of Education

Individual counselling and group classes were tailored to address and be considerate of the cultural customs, attitudes, values, norms, beliefs and literacy level of the Portuguese community. Portuguese-speaking dietician and nurse educators who were knowledgeable in the Portuguese culture facilitate both counselling and classes. Family members were strongly encouraged to accompany patients, particularly patients' spouses who are primarily responsible for household cooking and grocery shopping. There was a predominant use of visual aids, low literacy level education resources, and hands-on interactive activities that highlight traditional Portuguese foods, practices and customs in the group education (see Table 2).

The DEC had previously assessed the cultural appropriateness of the education services through patient focus groups facilitated by an experienced moderator not affiliated with the centre or patient population. The findings suggest that patients were able to understand and read the culturally specific nutrition literature and teaching tools offered by the centre. Patients enjoyed the use of traditional foods during cooking demonstrations and recipe modification exercises. They also identified situational problem-solving techniques as an effective means of learning to cope with challenging cultural situations such as eating at familial, social and celebratory events. Patients were willing to travel great distances to access these services in their language because it made them feel 'more comfortable'. Many of them described the group classes as having a 'family' or 'home away from home' feel. Patients felt that their cultural values and practices were taken into consideration in the delivery of the program and valued the staff that spoke their language and understood their culture. This report concluded that the program created a conducive learning environment for minority patients for whom language and literacy are often a barrier to learning and accessing services.

Individual Counselling Intervention

Portuguese-speaking educators provided initial assessments and ongoing individual follow-up visits for each Portuguese patient. With each patient, mutually agreed upon management goals and nutritional care plans were established with the educator during counselling. Also, throughout counselling, nutrition therapy is based on an assessment of individuals' metabolic profile (i.e. blood glucose, lipid and blood pressure) and on existing comorbidities such as renal nephropathy or gastro-intestinal complications as per evidence-based practice guidelines. At each visit, the nurse and dietician educators reassessed patients' status and identified patient priorities. The frequency and intensity (length of time) of each follow-up visit depended upon the patient's progress and achievement of their metabolic targets and educational objectives in accordance with Canada's standards for diabetes education. The most common course for follow-up is face-to-face communication with the same diabetes educator. Refer to Table 1 for a detailed description of the individual assessment and counselling interventions.

Group Education Classes

A multidisciplinary health care team facilitated group education (15 hours) over three consecutive weekdays. Group classes included approximately five to eight patients. The various teaching methods included didactic methods, mutual goal setting, situational problem solving, cognitive reframing and role-playing methods. See Table 2 for a detailed description of the group education program. One of the key components of group education is nutrition therapy (six hours). The nutrition component is predominantly interactive and sensory-stimulating. Throughout the nutrition classes, the dietician emphasized four key messages: (1) a limited and consistent intake of carbohydrates at each meal; (2) an adequate daily intake of fruits and vegetables; (3) lower intake of saturated fats; and (4) reduced fat in cooking. These are based on the nutrition therap\y guidelines for the management of diabetes. These key messages were also emphasized in one-on-one counselling. Both the individual and group education programs met the Standards for Diabetes Education in Canada.

Study Population

Participants were recruited over a two-year period between November 2001 and 2003. The DEC's clerk identified all newly referred Portuguese-speaking patients. This list was made available to the Portuguese-speaking research assistant who then called patients prior to their first scheduled appointment to inform them of the study.

Table 1 Individual Counselling Intervention[a]

Duration (min)	Content	Provider	Counselling method	Teaching tools
20–30	• Assess patient's signs and symptoms of hypoglycemia and presence of hyperglycemia. • Assess medical history: identify modifiable risk factors (hypertension, smoking, dyslipidemia, etc.). • Assess diabetes education history; determine diabetes education needs. • Assess financial constraints, coverage of medications, medical costs, patient concerns/fears	Dietician or nurse	• Teach treatment and prevention of hypoglycemia. • Refer to made to psychologist and/or social worker intervention as needed. • Orient to program and services.	Handouts
40–45	• Assess weight and weight history. • Assess diet history, factors affecting food intake, cooking and grocery shopping, ethnic foods preferred, fasting practices, diet history assessment and adequacy of nutritional intake.	Dietician	• Establish mutually agreed upon and realistic weight goals. • Set mutually agreed upon goals to modify diet and lifestyle. • Develop and instruct on individualized meal plan.	Handouts, food models for assessment of food portions, common food portion measuring tools.
30–45	• Assess medical history, medications and education needs. • Measure blood pressure and capillary blood glucose. Examine feet. • Assess self-monitoring of blood glucose (SMBG) technique or interest in SMBG. • Discuss insulin therapy, e.g. examination of injection sites.	Nurse	• Teach metabolic targets: blood pressure, blood glucose and serum lipids. • SMBG training. • Train self-administration of insulin for those starting on insulin therapy.	Handouts, glucose meters, teaching models.

[a]Counselling sessions are held face to face, involve family members or caregivers. Individual follow-up appointments are scheduled on a need basis per person.

Table 2 Group Education Program[a]

Duration (min)	Content	Provider	Teaching method	Teaching tools
Day 1 (08:30–15:00 h)				
75	Introductions: What is diabetes?	Nurse	Interactive, didactic	Flip chart, white board
75	Healthy eating and diabetes	Dietician	Interactive, didactic	Flip chart, white board, pictographic posters, food models and real food samples of culturally specific foods
60	Eating for a healthy heart	Dietician	Didactic, problem solving	Fat display showing fat content of food, problem solving and cognitive reframing
60	Diabetes and medication—oral medicines and over the counter medicines	Pharmacist	Didactic	Overheads and sample products/labels
Day 2 (08:30–15:00 h)				
75	High and low blood sugars	Nurse	Didactic, situational problem solving	Pictographic posters
	Treatment of low blood sugars	Nurse		
75	Meal planning	Dietician	Interactive, hands-on workshop and goal setting	Flip chart, real food samples, common measuring tools and typical plates/serving utensils
45	Cooking demonstration	Dietician	Interactive, hands-on workshop: a culturally traditional recipe is modified	Kitchen demonstration with specific ethnic food and ingredients
60	Diabetes and exercise	Registered phy-siotherapist	Didactic, demonstration of examples of warm-up, cool-down and some safe exercises demonstrated	Resistance stretch bands
60	Diabetes and medication—insulins	Pharmacist	Didactic	Graphic poster
Day 3 (08:30–15:30 h)				
75	Complications of diabetes	Nurse	Didactic	Teaching models (eyeball, heart, foot, kidney and atherosclerotic blood vessel)

Table 2 (*Continued*)

Duration (min)	Content	Provider	Teaching method	Teaching tools
75	Reading food labels, traditional events, special occasions and eating out	Dietician	Interactive, hands-on, role playing and situational problem solving	Food product labels, flip chart
60	Diabetes and stress, coping with diabetes	Psychologist	Interactive, sitting relaxation breathing exercise	
60	Diabetes resources	Social worker	Didactic	
Time remaining	Capillary blood glucose testing; weighing in, lunch and other breaks between classes[b]	Nurse and dietician		

[a]Education is provided face to face, supported by written literature in Portuguese (grade five literacy level). Lunch hour and breaks between classes not shown.

[b]Patients are weighed at the start of the group program. Capillary blood glucose testing occurs at a fasting state, before lunch and two hours after lunch each day of the three-day program. Patients share breakfast prior to beginning of classes on each day (for 20–30 minutes).

Patients who met the inclusion criteria, and who agreed to participate in the study, were asked to arrive at the centre a half-hour before their appointment. The inclusion criteria included diagnoses of type 2 diabetes, speaking Portuguese, willingness to participate in the Portuguese education programming and to be randomized into the intervention or control group. Exclusion criteria included being on renal dialysis, prior attendance at a similar educational program or the diagnosis of a mental illness. See Figure 1 for further details on exclusion and ineligibility criteria. The institution's research ethics board approved the study protocol.

Study Design

A randomized controlled trial (RCT) design was used to examine and compare the impact of the two groups (individual counselling or individual counselling plus group education) on nutrition adherence and glycemic control (HbA_{1c}) over a three-month period. A three-month interval for the RCT was chosen based on Franz *et al*. (2000), who found benefits from nutrition intervention after 6–18 weeks. Once study participants provided informed consent, the Portuguese-speaking research assistant administered the questionnaire. The research assistant also escorted participants to the lab for HbA_{1c} testing after their clinic appointment. Participants were randomly assigned (generated random number list) to either receive (1) individual counselling or (2) individual counselling with group education. A total of 87 participants were

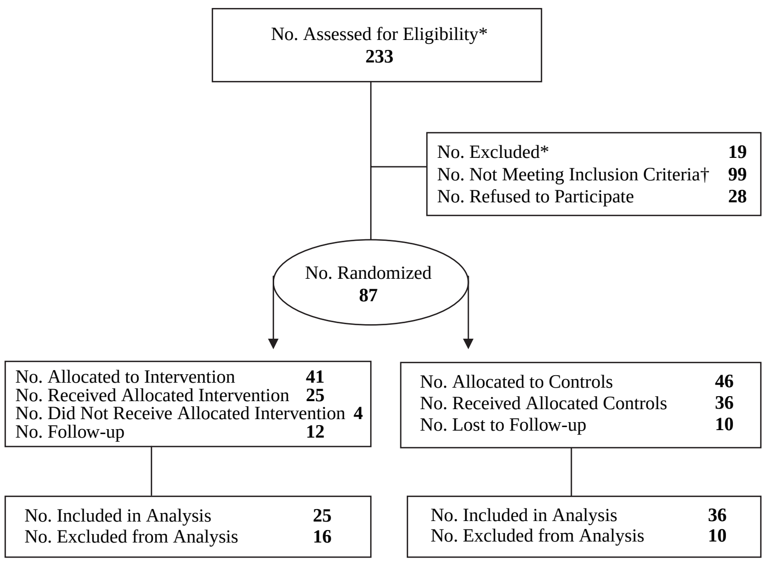

Figure 1 Recruitment Procedures.

*In eligibility: Patients were not available to attend the full three days of classes (work conflict); going on vacation within the next three months; unable to be reached prior to their scheduled appointment or did not attend their scheduled appointment therefore unable to recruit.

† Patients previously attended the DEC, were not of Portuguese ethnicity; claimed not to have diabetes; were not willing to sign consent form; or not willing to take blood test.

randomly assigned to individual counselling ($n = 46$) or individual counselling with group education ($n = 41$). However, only 61 participants completed the study requirements, 36 in the counselling only arm and 25 in the counselling and group education arm.

Appointments for counselling sessions and group classes were made for all participants by the DEC clerk. At three months, all participants were asked to come into the centre and complete the post-follow-up questionnaire and perform another HbA_{1c} test. Therefore, all participants completed questionnaires and performed an HbA_{1c} at the time of their first appointment and again, at three months after their first appointment. The research assistants were blinded to participants' intervention status. DEC providers were also blinded to patients' research participation status and were caring for all the participants regardless of the intervention assignment. Pharmacological treatment was not adjusted by DEC staff, but was managed by the participant's physician; therefore, we did not assess changes in pharmacological treatment.

By means of questionnaire, we also collected information such as age, sex, marital status, primary daily activity, education, income, and attitudes, subjective norms, perceived behavioural control, intention and self-reported adherence to nutrition recommendations. Family history of diabetes, the number of diabetes symptoms, medical management, body mass index and number of follow-up visits with either a nurse or dietician educator were extracted from the patient's medical chart after their first visit. Glycemic control (HbA_{1c} lab tests) was performed at the laboratory across the street from the centre.

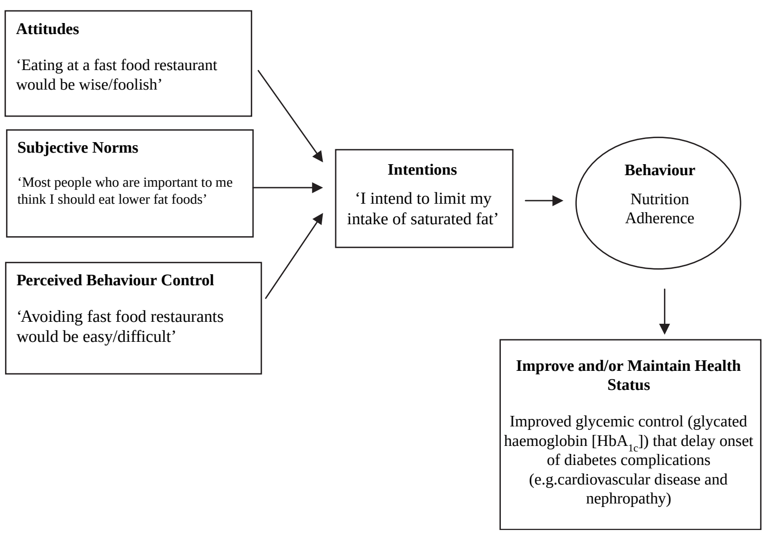

Figure 2 Theory of Planned Behaviour Framework for Nutrition Adherence and Glycemic Control.

Theoretical Framework

Language, literacy and cultural appropriateness are key elements in educational interventions for non-English-speaking populations. However, educational interventions need to go beyond merely being culturally competent. We still need to understand how interventions affect change, to identify the mechanisms through which self-care behaviours are shaped and maintained. Identifying the nature and influence of these mechanisms can determine the cause and effect of interventions. The Theory of Planned Behaviour (TPB) is a conceptual framework that predicts deliberate behaviour (Ajzen 1991). According to TPB, a person's intention to perform the behaviour (e.g. 'I intend to limit my intake of saturated fat') is a key predictor of that behaviour. Intentions to perform the behaviour are further determined by three components: attitudes, subjective norms and perceived behavioural control (PBC). Attitudes toward the behaviour refer to people's positive or negative evaluation of them performing the behaviour (e.g. 'Eating at fast food restaurants would be wise/foolish'). Subjective norms refer to people's perceptions of approval or disapproval from significant others for performing the behaviour (e.g. 'Most people who are important to me think I should eat lower fat foods'). Finally, PBC refers to people's judgment of their ability to perform the behaviour (e.g. 'Avoiding fast food restaurants would be easy/difficult') and is closely related to Bandura's (1977) concept of self-efficacy.

The more positive people's attitudes and subjective norms are regarding the behaviour, and the greater PBC over the behaviour, the more likely it is that people will intend to perform, and actually perform, the behaviour (see Figure 1). The literature suggests that measures of intention on average account for 34% of the variance in health-related behaviours (Godin & Kok 1996). This theory has been applied to a number of food-related behaviours such as food choice (Sparks 1994), reduction in fat intake (Paisley & Sparks 1998) and healthy eating (Povey *et al*. 1999). However, it has yet been applied to nutrition adherence in a diabetes population. In fact, the use of theoretical frameworks to understand the mechanisms through which interventions affect behavioural change, have not been extensively evaluated in the diabetes literature. This pilot tested the constructs of the TPB on nutrition management. We anticipate that examining its constructs will provide some insight into the mechanisms involved in adherence to nutrition management and potentially justify changes to future educational interventions.

Scales

Our questionnaire included two measures: (1) four adapted TPB scales on nutrition management for persons with diabetes and (2) a validated self-reported nutrition adherence scale for diabetes self-management (the Summary of Diabetes Self-care Activities Questionnaire). The questionnaire was translated and validated in Portuguese by a licensed bilingual interpreter and three Portuguese-speaking DEC staff members. The reading level was adjusted to be appropriate for the literacy level of the population. The final translation was agreed upon by group consensus over a period of several meetings. The questionnaire was piloted on 20 Portuguese-speaking patients with diabetes from the centre (but not part of the study sample). A Portuguese-speaking interviewer administered the questionnaire verbally to participants. Patients were asked to comment on the ease, comprehension and content of the questions. Based on comments, questions were modified to reduce redundancy and ambiguity and to better reflect the constructs of interest. We also increased the font size of the questionnaire to 14 inches. We placed all item responses on a large cardboard strip to facilitate ease of scoring. Overall, the questionnaire was well received and easy for patients to understand.

Outcome Variables

Theory of Planned Behaviour scales

The TPB scales consisted of questions pertaining to subjects' attitudes, subjective norms, PBC and intentions towards nutrition adherence. These scales were adapted from Sheeran and Orbell's use of TPB constructs for an exercise intervention (Sheeran & Orbell 2000). The structure, stem and response options were adopted, but our content focused on the nutritional targets for patients with diabetes: (1) a limited and consistent intake of carbohydrates at each meal; (2) an adequate daily intake of fruits and vegetables; (3) lower intake of saturated fats; and (4) reduced fat in cooking. Each scale consisted of eight items and were all measured on a five-point scale. Items were summed and a mean score calculated for each scale.

Responses for the attitudes scale ranged from 'good' to 'bad' or 'enjoyable' to 'unenjoyable' and had an internal consistency (IC) of Cronbach's $\alpha = 0.61$. Responses for the subjective norms scale ranged from 'strongly disagree' to 'strongly agree' or 'approve' to 'disapprove' and had an IC of $\alpha = 0.71$. Responses for the PBC control scale ranged from 'strongly disagree' to 'strongly agree' and had an IC of $\alpha = 0.84$. Responses for the intention scale ranged from 'strongly disagree' to 'strongly agree' or 'definitely do' to 'definitely do not' and had an IC $\alpha = 0.57$. These measures also have good construct validity as demonstrated by our analyses and reported in our result section. Please see the Appendix for TPB scales.

Adherence to nutrition management

The nutrition component of the Summary of Diabetes Self-care Activities Questionnaire (SDCA) (Toobert *et al*. 2000) was used to assess self-reported nutrition adherence for diabetes self-management. The nutrition component is measured on a five-point scale with responses ranging from 'never' to 'always'. Scale items ask for the percentage of times over the past seven days that respondents: followed a diet as recommended for diabetes; limited the number of calories as recommended; ate high fibre foods; ate high fat foods; and ate sweets and dessert. A higher score represents better adherence to the dietary recommendations for people with diabetes. The SDCA is a widely used self-report instrument for measuring diabetes self-management activities in adults. SDCA for the most part has adequate internal and test–retest reliability, evidence of validity (correlation with self-monitoring and interviews assessing similar behaviours) and sensitivity for change (Toobert *et al*. 2000). The internal consistency for the nutrition component of the SDCA was $\alpha = 0.57$ in our study.

Glycemic control

Glycemic control, a widely used term in the field of diabetes management, describes the degree to which both self-monitoring blood glucose levels and glycosylated hemoglobin A_{1c} levels are compatible with recommended levels defined in the clinical practice guidelines for the management of diabetes. The recommended levels are associated with reduced development or progression of diabetes complications. Glycosylated hemoglobin A_{1c} (HbA_{1c}) is a physiological indicator of blood glucose in the blood over the previous three months from day of testing (Mayer & Freedman 1983). HbA_{1c} is recognized as the most useful clinical indicator of glycemic control for patients with diabetes. Because the morbidity and mortality associated with diabetes are so profound, any improvement in glycemic control has the potential to enhance the quantity and quality of life for persons with diabetes (UK Prospective Diabetes Study Group 1998). The goal of diabetes self-management is to keep glycemic control levels below 7% (Canadian Diabetes Association Clinical Practice Guidelines Expert Committee 2003).

Statistical Analysis

Since the goal of our study was to explore the efficacy of two education delivery methods, only study participants who completed their assigned intervention and the questionnaires were included in the analyses (i.e. per-protocol analyses) (Roland & Torgerson 1998). Demographic variables were analysed using Pearson's chi-square statistic for categorical variables and independent t-tests or Mann–Whitney test for continuous variables. A paired t-test was used within each group to assess change from pre- to post-test scores. Analysis of covariance (Twisk & Proper 2004), with the corresponding pre-test score as the covariant, was performed to assess between group differences in outcomes (TPB constructs, nutrition adherence and glycemic control). Two sample t-tests were performed to describe absolute mean difference from pre- to post-test scores between groups.

To test construct (convergent) validity of the TPB scales, Pearson's correlations were used to test the relationship between TPB constructs (independent variables) and nutrition adherence and glycemic control (dependent variables). Three multiple linear regression models were examined: (1) attitudes, subjective norms and PBC on intentions; (2) attitudes, subjective norms, PBC and intentions on nutrition adherence; and (3) attitudes, subjective norms, PBC, intentions and nutrition adherence on glycemic control.

Descriptive statistics were examined to verify that the statistical assumptions required for each method of analysis were satisfied. Multicollinearity was not an issue in our multivariate regression model. The level of statistical significance was $p < 0.05$ with two-sided tests. SPSS software (release 12, 2003, SPSS Inc.) was used for all analyses.

Results

Participants' Characteristics

The main reasons non-participants gave for refusing to take part in the study included work conflict, and lack of time and interest in study participation. The need for participants to commit themselves to three full consecutive days of classes affected recruitment (see Figure 1). Study participants had more diabetes symptoms at the time of their first visit than non-participants (2.60 ± 1.6 vs 1.71 ± 1.4, $p < 0.05$). The response rate for the study was 76%. The difference between study completers and non-completers was that non-completers were more likely to be unemployed at baseline (data not shown). Possibly, after agreeing to participate in the study some unemployed participants might have found jobs that precluded them from adhering to the study requirements, particularly in the counselling plus group education intervention.

Descriptive characteristics of the study population are summarized in Table 3. The study sample was more likely to be females, married, have less than a grade nine education, a gross personal income of less than $19,000 and have a family history of diabetes. Baseline scores for TPB scales (not shown) did not differ significantly between intervention groups. In addition, there were no differences in the number of individual follow-up appointments between the two groups over the three-month period.

Intervention Effects

Regardless of education format, all study participants significantly improved from pre-intervention to post-intervention in attitudes (change ($\triangle$) = 2.28), subjective norms ($\triangle = 0.43$), PBC ($\triangle = 0.37$) and intentions to adhere to nutrition recommendations ($\triangle = 0.37$), self-reported nutrition adherence ($\triangle = 0.39$) and HbA_{1c} ($\triangle = -0.51\%$). HbA_{1c} decreased from $7.4\% \pm 1.6$ at baseline to $6.9\% \pm 1.3$ at three months ($p < 0.01$) (see Table 4).

Our results also indicated that participants receiving individual counselling in conjunction with group education had significantly greater PBC, more positive intentions and better adherence toward nutrition management at three months than those who only received individual counselling. In addition, we observed trends in the hypothesized direction for attitudes and subjective norms ($p = 0.09$ and 0.06, respectively). Although improvements in TPB constructs were marginally better in

Table 3 Characteristics of Study Population ($N = 61$)

Characteristic	Individual ($n = 36$) N (%)/mean ± SD	Group ($n = 25$) N (%)/mean ± SD	p-Value
Age	59.0 ± 12.1	60.4 ± 7.92	0.58
Sex			0.91
Male	11 (30.6)	8 (32.0)	
Female	25 (69.4)	17 (68.0)	
Marital status			0.14
Married	26 (72.2)	22 (88.0)	
Separated/divorced/ widowed	10 (27.8)	3 (12.0)	
Primary daily activity[a]			0.63
Employed	8 (22.2)	3 (12.5)	
Unemployed	15 (41.7)	11 (45.8)	
Retired	13 (36.1)	10 (41.7)	
Education			0.69
< Grade 9	32 (88.9)	23 (92.0)	
≥ Grade 9	4 (11.1)	2 (8.0)	
Gross personal income[b]			0.53
≤ $19,999	28 (84.8)	18 (78.3)	
> $19,999	5 (15.2)	5 (21.7)	
Family history of diabetes			0.74
Yes	29 (80.6)	18 (72.0)	
No	3 (8.3)	3 (12.0)	
Not known	4 (11.1)	4 (16.0)	
Length of diagnosis (years)[c]	1 ± 1–7.5	4 ± 1–12.5	0.24
Total number of DM symptoms	2.6 ± 1.5	2.7 ± 1.6	0.81
Diabetes management			0.32
Diet only	11 (30.6)	4 (16.0)	
Oral agents	24 (66.7)	19 (76.0)	
Insulin and oral agents	1 (2.8)	2 (8.0)	
Self-monitoring of Blood Glucose			0.80
Yes	17 (47.2)	11 ± 44.0	
No	19 (52.8)	14 ± 56.0	
Baseline HbA_{1c}	7.0 ± 0.07	7.0 ± 0.02	0.96
Weight (kg)	87.8 ± 19.4	85.3 ± 12.6	0.55
Height (cm)	158.0 ± 8.3	157.0 ± 8.6	0.60
BMI (kg/m)	35.0 ± 6.6	34.9 ± 5.6	0.91
Number of individual follow-up visits	1.83 ± 0.69	2.08 ± 0.95	0.25

[a]One (4.0%) subject in the experimental group did not provide data for this item.

[b]Three (8.3%) subjects in the control group and two (8.0%) in the experimental group did not provide data for this item.

[c]Median and interquartile range; Mann–Whitney test used to test difference in medians.

participants receiving the intervention, no significant differences were observed for HbA_{1c} between the two groups. The proportion of well controlled subjects ($HbA_{1c} < 7.0\%$), receiving individual counselling and individual counselling plus group education pre-intervention was 58 and 52%, respectively (not statistically different). By the end of the study period, the proportions increased for both groups to 64 and 72%, but were still not statistically different.

Relationship among TPB Model Components

Correlations among the various model constructs at three months are presented in Table 5. Attitudes ($r = 0.345$), subjective norms ($r = 0.339$) and PBC ($r = 0.413$) were significantly and positively correlated with intentions to adhere to nutrition recommendations and with each other. To determine how well these variables simultaneously predict intentions, a multivariate linear regression analysis was conducted. Attitudes, subjective norms and PBC explained 18.1% of the variance in intentions to adhere to nutrition recommendations with only PBC ($\beta = 0.296$) making significant contributions (see Table 6).

Intentions ($r = 0.479$), subjective norms ($r = 0.430$), PBC ($r = 0.306$) and attitudes ($r = 0.245$) were significantly correlated with self-reported adherence to nutrition recommendations. In our regression model, attitude, subjective norm, PBC and intention explained 27.6% of the variance in nutrition adherence with only intentions ($\beta = 0.377$) and subjective norms ($\beta = 0.338$) making significant contributions (see Table 4).

Self-reported adherence to nutrition recommendations and the TPB constructs were not significantly correlated with glycemic control (HbA_{1c}) in both bivariate (see Table 4) and multivariate analyses (data not shown).

Discussion

Education Intervention

Our results provide supporting evidence that a culturally tailored and language-specific program for Portuguese patients with type 2 diabetes, either as individual counselling or individual counselling with group education, is efficacious in improving patient's nutrition adherence and glycemic control (HbA_{1c}) over a three-month period. HbA_{1c} decreased approximately 0.5% in the entire study cohort during the three-month period. Landmark studies demonstrate that a reduction of 0.5% in HbA_{1c} is associated with a significant reduction in diabetes-related complications (The Diabetes Control and Complications Trial Research Group 1993). Any reduction in hyperglycemia is shown to be beneficial (UKPDS 1998a, b). Hence, this reduction observed in our study is clinically significant. However, longitudinal research is needed to observe if this reduction in glycemic control is maintained over time.

Table 4 Behavioural, Nutritional and Clinical Outcomes by Two Education Formats

Measurement	Intervention format[a]						Overall cohort change[b]	Differences between intervention format			
	Individual only ($n=25$)			Individual plus group ($n=36$)				Post-test values adjusted for baseline[c]			Mean difference[d]
	Baseline	3 Months	Mean difference	Baseline	3 Months	Mean difference	Mean difference	Individual	Group	p-Value	
Attitudes	2.07 ± 0.09	4.23 ± 0.07	2.16 ± 0.13***	1.99 ± 0.11	4.43 ± 0.07	2.45 ± 0.13***	2.28 ± 0.10***	4.23 ± 0.07	4.42 ± 0.08	0.09	0.20 ± 0.11
Subjective norms	3.85 ± 0.10	4.18 ± 0.90	0.33 ± 0.10***	3.85 ± 0.10	4.43 ± 0.10	0.58 ± 0.13***	0.43 ± 0.08***	4.18 ± 0.08	4.42 ± 0.10	0.06	0.25 ± 0.13
PBC	3.76 ± 0.13	4.12 ± 0.07	0.36 ± 0.13*	3.90 ± 0.12	4.29 ± 0.09	0.38 ± 0.11**	0.37 ± 0.09***	4.13 ± 0.07	4.27 ± 0.07	0.03	0.16 ± 0.11*
Intentions	3.98 ± 0.07	4.27 ± 0.76	0.28 ± 0.08***	3.97 ± 0.08	4.47 ± 0.08	0.51 ± 0.06***	0.37 ± 0.05***	4.26 ± 0.06	4.48 ± 0.07	0.03	0.20 ± 0.11*
Nutrition adherence	3.58 ± 0.10	3.78 ± 0.10	0.20 ± 0.08**	3.54 ± 0.12	4.20 ± 0.10	0.66 ± 0.15***	0.39 ± 0.08***	3.77 ± 0.08	4.21 ± 0.10	0.001	0.42 ± 0.14*
HbA_{1c}	7.38 ± 0.27	6.88 ± 0.23	−0.49 ± 0.15***	7.36 ± 0.32	6.80 ± 0.24	−0.54 ± 0.19**	−0.51 ± 0.90***	6.88 ± 1.25	6.81 ± 1.50	0.75	0.08 ± 0.34

[a]Paired t-test with means ± standard error of the mean.
[b]Two-sample t-test to compare pre-test to post-test in entire cohort with means ± standard error of the mean.
[c]Analysis of covariance to compare group and control post-test scores, adjusting for pre-test score with means ± standard error of the mean.
[d]Two-sample t-test to describe pre-test to post-test mean change scores between groups with mean ± standard error of difference between the means.
*$p<0.05$, **$p<0.01$, ***$p<0.001$.

Table 5 Correlation Matrix between TPB Constructs, Nutrition Adherence and Glycemic Control at Three Months

	1	2	3	4	5	6
1. Intentions		0.345**	0.339**	0.413**	0.479**	−0.193
2. Attitudes			0.574**	0.436**	0.245**	−0.197
3. Subjective norms				0.413**	0.430**	−0.118
4. Perceived behavioural control					0.306**	−0.008
5. Nutrition adherence						−0.136
6. HbA_{1c}						

Note: *$p < 0.05$, **$p < 0.01$.

Educational interventions that target cultural values, beliefs and language barriers of a specific ethnic group are known to increase patient communication, patient satisfaction with care providers and satisfaction with their health care experiences (Collins *et al*. 2002). Furthermore, culturally competent DSME are far more effective in retaining non-English-speaking patients (Anderson *et al*. 1991). By tailoring education according to culture and language, DECs can potentially provide more effective management programs.

Our findings also suggest that both counselling and counselling with group education improve attitudes, subjective norms, PBC, intentions and adherence to nutrition management. Even though these improvements are observed in all participants, greater improvements are seen in those receiving individual plus group education at three months. Nevertheless, glycemic control remains similar in the two

Table 6 Regression of Intentions, Nutrition Adherence and Glycemic Control at Three Months

Predictors	Adjusted R^2	Adjusted R^2 change per variable	B	SE B	β
Outcome: Intention	0.18				
Step 1. Attitude	0.11	0.11	0.141	0.143	0.136
Step 2. Subjective norm	0.12	0.01	0.112	0.110	0.139
Step 3. Perceived behavioural control	0.18	0.06	0.289	0.121	0.296*
$F(3,66) = 6.07$***					
Outcome: Nutrition adherence	0.28				
Step 1. Attitudes	0.05	0.05	−0.146	0.186	−0.103
Step 2. Subjective norm	0.16	0.11	0.373	0.143	0.338*
Step 3. Perceived behavioural control	0.17	0.01	0.075	0.163	0.056
Step 4. Intentions	0.28	0.11	0.516	0.159	0.377**
$F(4,65) = 7.56$***					

Note: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, B = beta; SE B = standard error of beta, β = standardized beta.

groups. This observation may be explained by the fact that half of our study population had fairly good glycemic control at baseline, thereby limiting the power to detect small but clinically important improvements in glycemic control between groups.

While the literature suggests that group learning provides sound educational advantages and helps adults reach a greater degree of diabetes self-management than in an individual learning environment (Raz *et al*. 1988; Trento *et al*. 2001, 2002; Rickheim *et al*. 2002), a larger RCT with a longer follow-up period is in order to evaluate the long-term effects of counselling only and counselling with group education in a Portuguese population.

Theory of Planned Behaviour

Using the TPB provided some insight into the mechanisms involved in adherence to nutrition recommendations. An important finding was that PBC, also known as self-efficacy, is significantly associated with intentions to adhere to nutrition recommendations. Although indirectly associated with nutrition adherence in our study, self-efficacy is a consistent predictor of behavioural outcomes in diabetes management. Several studies demonstrate that a higher sense of self-efficacy in diabetes management is associated with higher levels of adherence to diabetes self-care behaviours and glycemic control (Howorka *et al*. 2000). Educational components that target self-efficacy such as reinforcement, observational learning, goal setting and problem solving may be necessary to improve dietary adherence.

Another important finding was that intentions as well as subjective norms are associated with dietary adherence. These findings propose that eating behaviours remain largely within one's personal drive; however, the inability to enact one's intention in some areas may result from various environmental factors. Behavioural practices such as dietary management are subjected to social normative influences (Bandura 1977); for instance, family members are typically involved in buying and preparing foods. The literature strongly supports the association between higher levels of perceived social support and better adherence to diabetes self-care behaviours (Fukunishi *et al*. 1998). Involving family members in the education process may better support patients to successfully adhere to nutrition recommendations.

Self-reported adherence to nutrition recommendations did not reflect improved glycemic control in our study population. However, these observations are not uncommon in the literature considering that HbA_{1c} levels are also influenced by genetic and clinical factors other than diet (Jeffcoate 2004). In any event, our results underscore the importance of measuring a diversity of health processes and outcomes that may provide information on various aspects of nutrition adherence, rather than focusing solely on physiological indicators.

Limitations

Given the fact that this was a pilot study in a culturally specific population, our study sample was small. Recruitment was difficult as a few patients were not available for randomization into the three-day group education, primarily due to work responsibilities. As such, many employed individuals were not eligible to participate, thus decreasing the generalizability of our results. This limitation is inherent in any intensive education intervention delivered during standard work hours. Nonetheless, we recommend that intensive group education be delivered in shorter intervals during the day or evenings. Moreover, our study follow-up period was only three months and therefore we were unable to assess maintenance of desired outcomes.

Another limitation to our study was our moderately low internal consistency scores for our scales. The low scores were probably due to the relatively small number of items (eight) for each scale (Crockers & Algina 1986). Given that our study population are non-English-speaking seniors who are not familiar with cognitive style questions, the authors considered that slightly lower reliability estimates were more desirable than an excessively long and burdensome instrument (Mehrens & Lehman 1991). Additionally, our scales were tapping into heterogeneous behaviours (different aspects of diet) that normally decrease internal consistency. Under such circumstances, the importance of content validity outweighs internal consistency, since the ultimate goal of the instrument was inferential, which depends more on content validity.

Since we only conducted per-protocol analyses, study participants may have differed from those who dropped out in ways that are related to the outcome. Thus, our estimated effects may be biased in a conservative direction (i.e. acceptance of the null hypothesis of no difference). An intention-to-treat analysis was not conducted because we were primarily interested in the efficacy, and not the effectiveness, of the intervention at this pilot stage. This seems reasonable given that participants can only be affected by an intervention they actually received. Secondly, participants who dropped out of the study declined to provide further information; thus, we had incomplete data for those participants in both intervention arms and were unable to conduct intention-to-treat analyses.

Suggestions for future research include the use of a more thorough assessment of dietary intake such as a food frequency questionnaire or 24-hour dietary recall. Such methods may better detect a relationship between dietary adherence and glycemic control and be less subjected to social desirability. Future research should also assess the impact of education on other significant dietary-related factors such as dyslipidemia, hypertension and weight reduction. Lastly, future studies should assess and compare the cost-effectiveness of both individual education and individual in conjugation with group education.

Conclusions

In response to the increasing globalization and international migration of people, it is crucial that national health education initiatives increase their attention to the need for efficacious and effective culturally and linguistically competent programs. Our study findings provide some preliminary evidence that culturally competent education initiatives for Portuguese adults with type 2 diabetes may be efficacious in shaping eating behaviours. Future studies are needed to determine the long-term benefits of counselling and counselling in conjunction with group education.

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Appendix

Theory of Planned Behaviour scales adapted for nutritional targets in the management of diabetes.

1. Intentions

How much do you agree with the following statements:
Responses: (strongly disagree, disagree, neutral, agree, strongly agree)

How much do you intend to do the following behaviours?
Responses: (definitely do, 2 [no tag], neutral, 3 [no label], definitely do not)

a. I intend to maintain the same portion of bread, cereal, rice, pasta or potatoes at each meal.
b. I intend to eat fruit and/or vegetables at each meal.
c. I intend to minimize the amount of animal fats in my diet (lean meats, skinless poultry).
d. I intend to use little or no fats in cooking (e.g. margarine, butter, oil, olive oil or lard).

2. Attitudes

What are your attitudes toward the following health behaviours?
Responses: (good to bad [1, 2, 3, 4, 5]) and (enjoyable to not enjoyable [1, 2, 3, 4, 5])

a. Maintaining the same portion of bread, cereal, rice, pasta or potatoes at each meal would be:
b. Eating fruit and vegetables at every meal would be:
c. Limiting the amount of animal fats in my diet would be:
d. Using little or no fats in cooking would be:

3. Social Norms

Most people who are important to me think that I should:
Responses: (strongly disagree, disagree, neutral, agree, strongly agree)

People in my life whose opinion I value would approve of me ...
Responses: (approve, 2 [no tag], neutral, 3 [no tag], disapprove)

a. Maintain the same portion of bread, cereal, rice, pasta or potatoes at each meal.
b. Eat fruit and/or vegetables at each meal.
c. Minimize the amount of animal fats in my diet (lean meats, skinless poultry).
d. Use little or no fats in cooking (e.g. margarine, butter, oil, olive oil or lard).

4. Perceived Behavioural Control

If I wanted to, I could easily:
I feel complete control over whether or not I:
Responses: (strongly disagree, disagree, neutral, agree, strongly agree)

a. Maintain the same portion of bread, cereal, rice, pasta or potatoes at each meal.
b. Eat fruit and/or vegetables at each meal.
c. Minimize the amount of animal fats in my diet (lean meats, skinless poultry).
d. Use little or no fats in cooking (e.g. margarine, butter, oil, olive oil or lard).